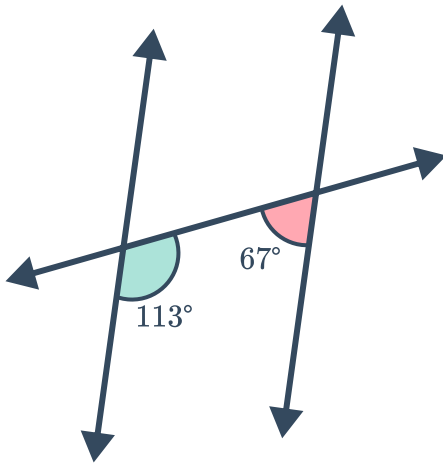


## Proving lines parallel

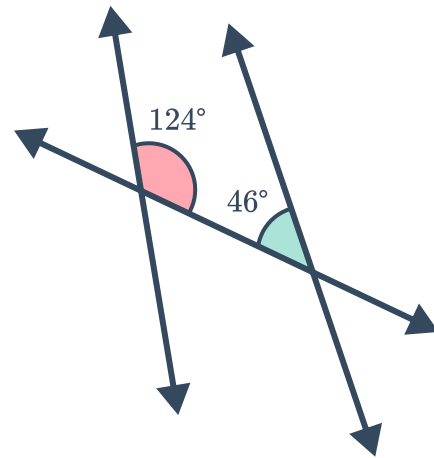


- 1 For the following, state whether the diagram contains parallel lines. Justify your choice with a theorem or postulate.

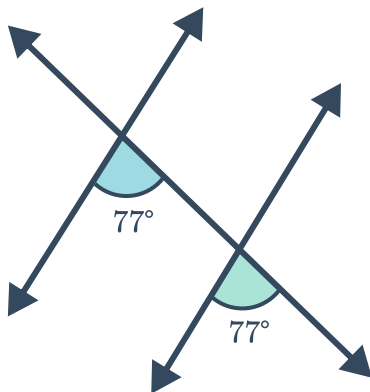
a



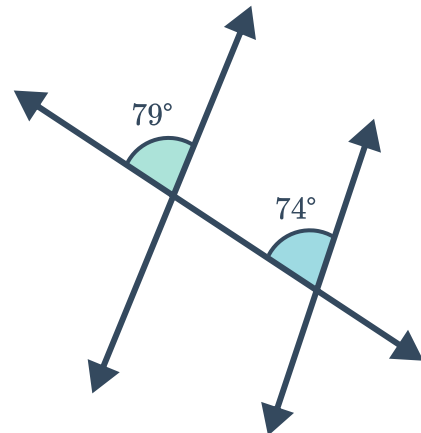
b



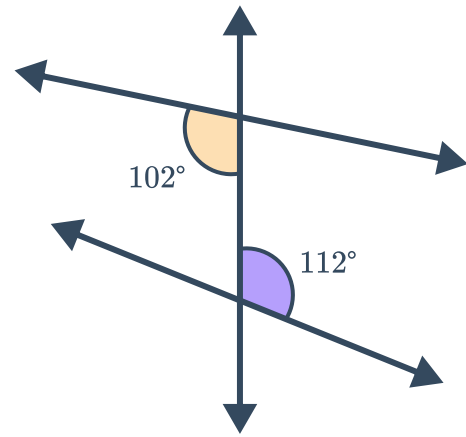
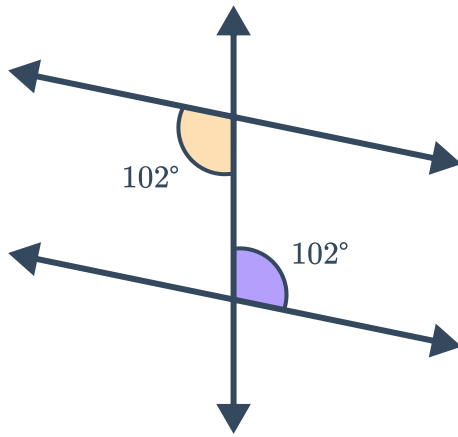
c



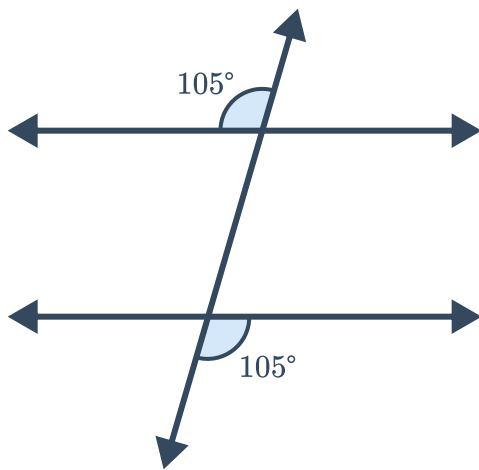
d



e

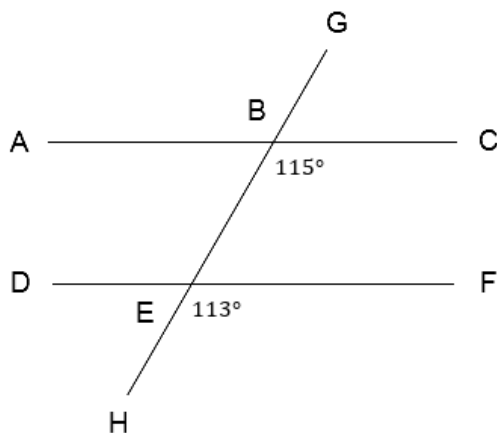


g

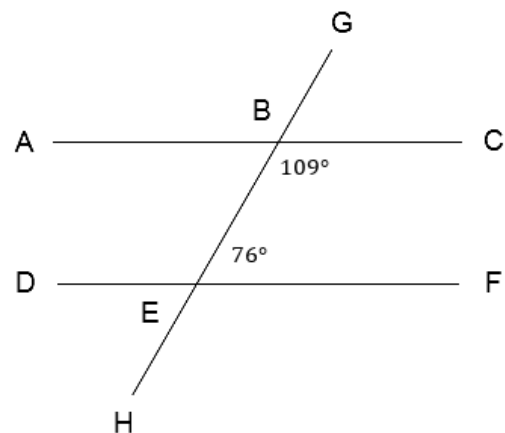


2 For the following diagrams, state whether  $\overline{AC}$  and  $\overline{DF}$  are parallel. Explain your reasoning.

a

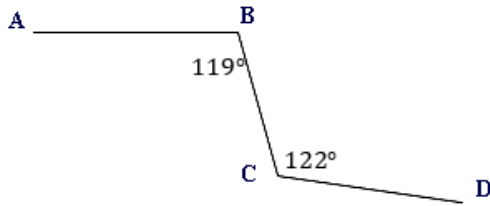


b

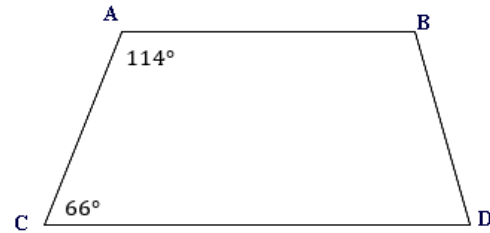


3 For the following diagrams, state whether  $\overline{AB}$  and  $\overline{CD}$  are parallel. Explain your reasoning.

a

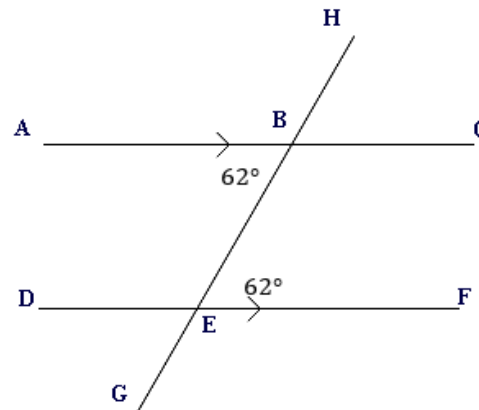


b



4 Eileen was given the following diagram, in which  $m\angle ABE = m\angle BEF = 62^\circ$ . She immediately marked the parallel symbols on the diagram to show that  $\overline{AB} \parallel \overline{DE}$ . State whether the following reasons justify Eileen's markings.

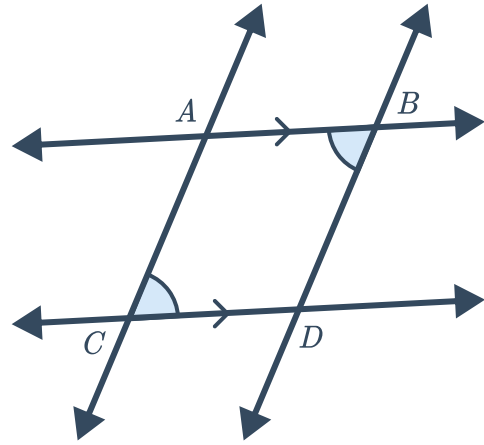
- The two congruent angles form a pair of corresponding angles. Since the corresponding angles are congruent,  $\overline{AB}$  must be parallel to  $\overline{DE}$ .
- The diagram is drawn to scale and so the lines look to be parallel.
- The two congruent angles form a pair of alternate interior angles. Since the alternate interior angles are congruent,  $\overline{AB}$  must be parallel to  $\overline{DE}$ .
- The two congruent angles form a pair of consecutive interior angles. Since the consecutive interior angles are congruent,  $\overline{AB}$  must be parallel to  $\overline{DE}$ .



5 Suppose that two lines,  $\overleftrightarrow{AB}$  and  $\overleftrightarrow{CD}$ , are intersected by a transversal. Which of the following reasons alone could be used to show that the lines are parallel?

- Consecutive interior angles converse
- Alternate exterior angles theorem
- Linear pair theorem
- Congruent supplements theorem

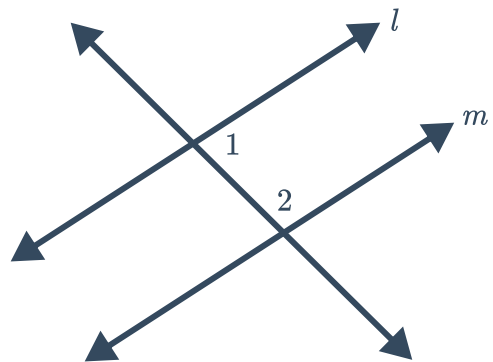
- 6 The following diagram shows that  $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$  and  $\angle ABD \cong \angle DCA$ .  
Prove that  $\overleftrightarrow{AC} \parallel \overleftrightarrow{BD}$ .



## Additional questions

- 7 Review the statement. Describe the error.

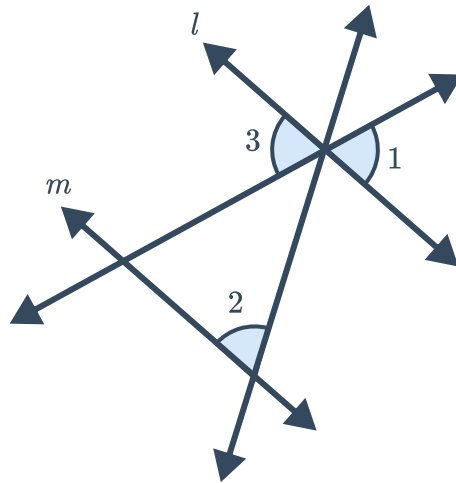
- If  $\angle 1 \cong \angle 2$ , then by the converse of the consecutive interior angles theorem,  $l \parallel m$ .



- 8 Review the following proof. Describe the error.

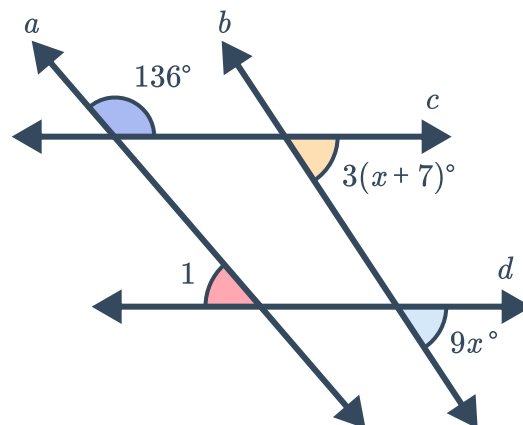


$\angle 1 \cong \angle 2$  is given. By the vertical angles theorem,  $\angle 1 \cong \angle 3$ . So by the transitive property of congruence,  $\angle 2 \cong \angle 3$ . By the converse of the corresponding angles theorem,  $l \parallel m$



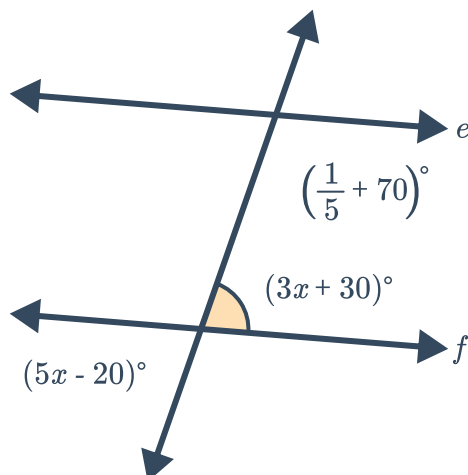
- 9 Consider the given diagram:

- Find the value of  $x$  such that  $c \parallel d$ .
- Find  $m\angle 1$  such that  $c \parallel d$ . Justify your reasoning.

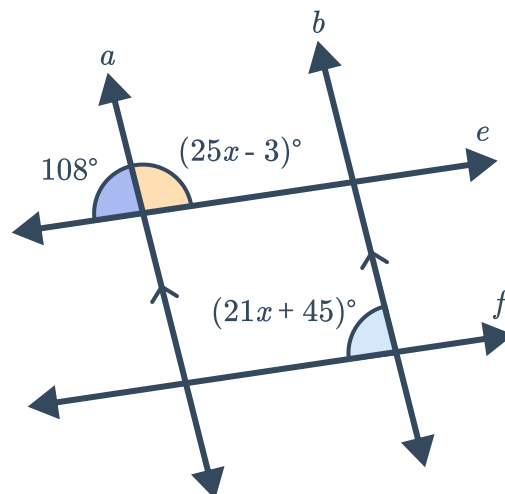


- 10 Determine the value of  $x$  that makes  $e \parallel f$ .

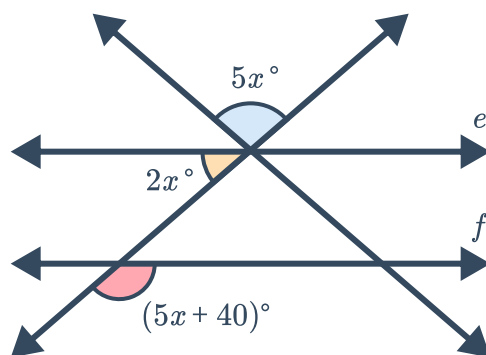
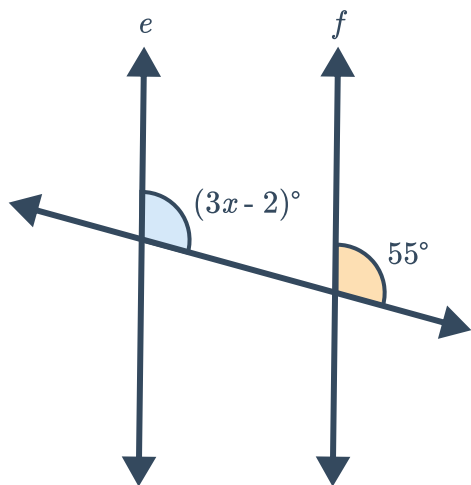
a



b



c



- 11 Consider the following diagram for the below statements. Determine if the information given is enough to justify the conclusion. Explain your reasoning.



a Given:

- $\angle 1$  and  $\angle 14$  are supplementary angles
- $\angle 22 \cong \angle 1$

Conclusion:  $a \parallel b$

b Given:

- $\angle 7$  and  $\angle 10$  are supplementary

Conclusion:  $b \parallel c$

c Given:

- $b \parallel c$
- $\angle 5$  and  $\angle 19$  are supplementary

Conclusion:  $a \parallel b$

d Given:

- $\angle 1$  and  $\angle 11$  are supplementary

Conclusion:  $a \parallel c$

e Given:

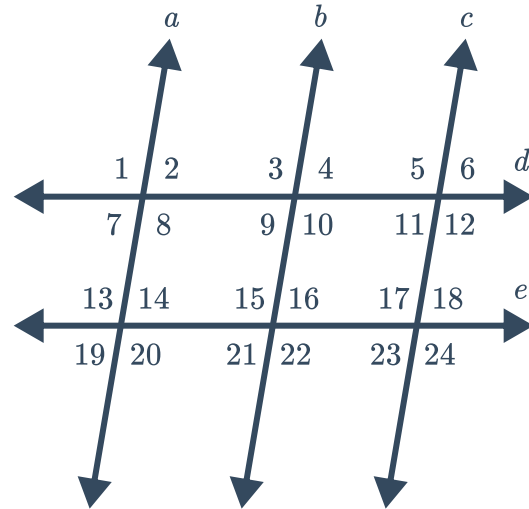
- $\angle 18$  and  $\angle 20$  are supplementary
- $a \parallel b$

Conclusion:  $b \parallel c$

f Given:

- $\angle 1 \cong \angle 13$
- $\angle 3 \cong \angle 24$

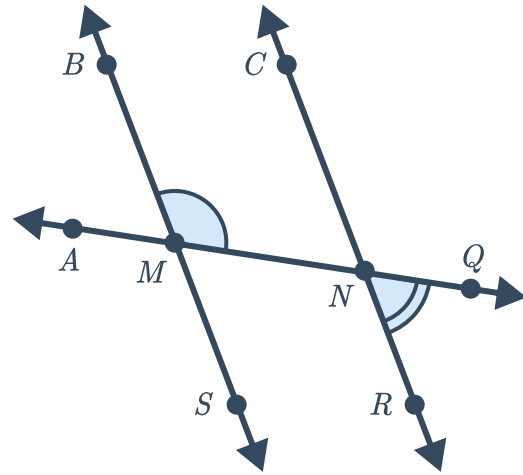
Conclusion:  $a \parallel b$



12 Consider the following diagram:



- a Identify the angles that are supplementary to  $\angle BMN$ .
- b If  $\angle BMN$  and  $\angle QNR$  are supplementary, prove  $\overleftrightarrow{BS} \parallel \overleftrightarrow{CR}$ .



13 Given:

- $\angle A$  and  $\angle D$  are supplementary
- $\angle B \cong \angle C$

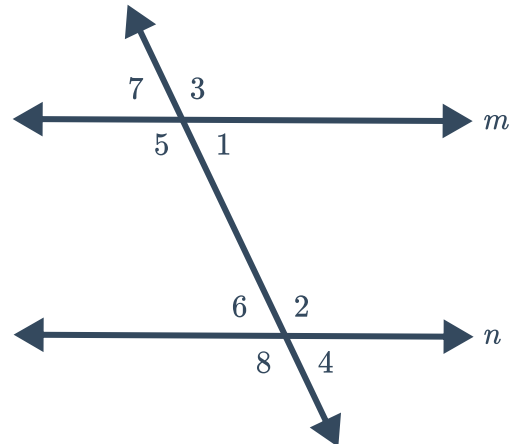
Prove:  $\overline{AD} \parallel \overline{BC}$



14 Prove the converse of the consecutive interior angles theorem:

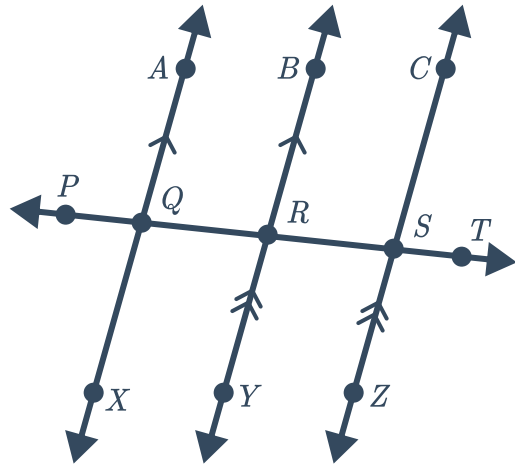
Given:  $\angle 1$  is supplementary to  $\angle 2$

Prove:  $m \parallel n$

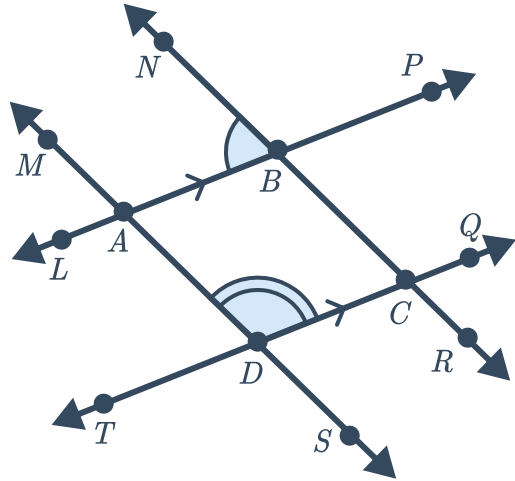




- 15 Given:  $\overleftrightarrow{AX} \parallel \overleftrightarrow{BY}$  and  $\overleftrightarrow{BY} \parallel \overleftrightarrow{CZ}$   
 Prove:  $\angle CST \cong \angle XQP$



- 16 Given:  $\overleftrightarrow{AB} \parallel \overleftrightarrow{DC}$  and  $\angle NBA$  and  $\angle ADC$   
 are supplementary  
 Prove:  $\overleftrightarrow{AD} \parallel \overleftrightarrow{BC}$



- 17 Given:  $\overleftrightarrow{AB} \parallel \overleftrightarrow{DC}$  and  $\angle MAB \cong \angle BCQ$   
 Prove:  $\overleftrightarrow{AD} \parallel \overleftrightarrow{BC}$

